Escape speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain escape using a simplified gravitational-energy calculation.

SUCCESS CRITERIA - I can...

1. State the zero-at-infinity energy convention.
2. Distinguish escape speed from circular-orbit speed.
3. Name neglected effects in the ideal calculation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “escape speed” and “orbit” to explain your answer. Sentence starters are optional.

escape speed: minimum ideal speed to leave without further thrust

orbit: a path around another body under gravity

radius: distance from the center to the edge of a circle or sphere

01 Warm-up: bring back an earlier idea

A **3 kg** cart at **10 m/s** hits an identical cart at rest. Final velocities are **4** and **6 m/s**. **Check momentum** and energy; classify the collision.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

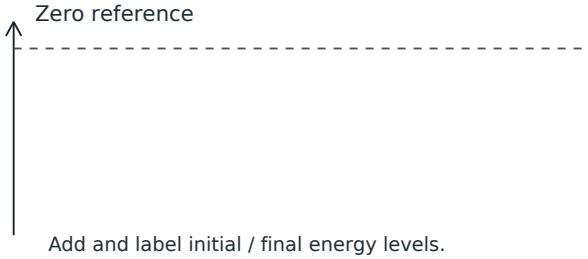
02 Worked example: explain the prediction

Reference: $v_{escape} = \sqrt{(2\ GM/r)}$; $v_{circular} = \sqrt{(GM/r)}$.
Ideal spherical body; ignore atmosphere/rotation. Minimum escape reaches infinity at zero speed with no further thrust.

Worked reasoning: For a simulated planet with GM=**50 m³/s²** and radius **4 m**, escape speed is $\sqrt{(2\ GM/R)}$=**5 m/s**. At that limiting speed the ideal final speed at infinity is zero.

Your turn: Draw initial kinetic and gravitational-potential-energy bars whose total is zero.

Sentence starter: In my work, I used … to show … . I labeled it … .



Add and label initial / final energy levels.

Label K/m, U/m and total E/m in J/kg.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Does escaping mean that gravity turns off at the launch point? **Explain what the speed allows** using energy calculations.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

04 Apply the physics

For GM=**400 m³/s²** and r=**8 m**, calculate circular speed and escape speed.
1. **Compare their ratio**.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

Escape speed

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why is an ideal escape-speed calculation not a complete rocket-launch plan?

Word help: ideal = using stated simplifying conditions

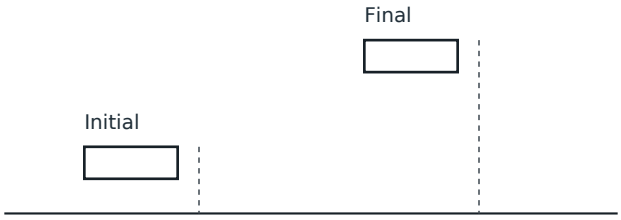
Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **6 kg** object descends **8 m** near Earth. Use g=**10**. Find ΔU and the work done by gravity.
Label the distances or energy levels used in your reasoning.

Sentence starter: I found … [unit]. This value means … in this situation.



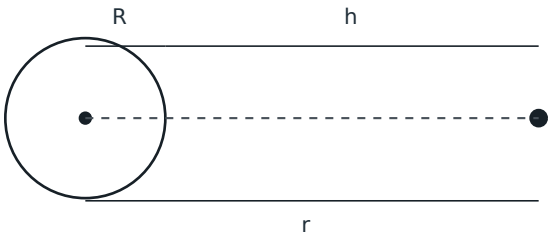
Choose and label a zero-height level.

Marzano check - circle: 1 2 3

07 Exit check

At fixed GM, launch distance increases by factor **4**.
1. **Compare escape speeds**.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

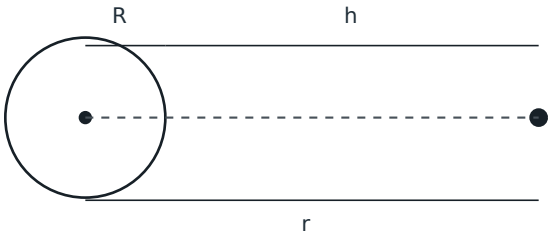


Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A planet's mass doubles while radius stays fixed. **Compare escape speed**. Then double both mass and radius and **compare again**. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3